Supporting Information: Route-Aware Computational Selection of Redox Motifs for Preformed Polymers

Working draft: authorship and affiliations to be confirmed

Evidence revision: 23 October 2026

Internal working draft. This SI contains finite method controls and implementation checks. Pending scientific analyses remain explicit.

1 Benchmark intervals and post hoc reference diagnostics

The complete panel has 39 endpoints from 30 molecules and seven studies. Molecule-cluster 95% percentile bootstrap intervals are [0.213,0.286] V for MAE, [0.245,0.313] V for RMSE and [0.128,0.257] V for bias. Study-cluster intervals are [0.178,0.306], [0.211,0.325] and [0.053,0.285] V, respectively. Both use 2,000 replicates and seed 20261021. The seven study clusters have unequal sizes; their percentile intervals are weakly calibrated. Molecule resampling retains common study and family effects and is also not a prediction interval for a new molecule.

For each cross-validation fold, a constant residual offset is estimated only from training endpoints and subtracted from the held-out prediction. Molecule and study folds remove all endpoints of the held-out molecule or DOI. The two-offset model fits separate constants for first and second redox steps. These choices were motivated by inspection of the complete retrospective panel and are internal diagnostics, not an untouched external test. Relative potentials to TEMPO,

Table S1: Potential errors (V); primary unfitted statistics and internal offset diagnostics.

Holdout	Offset model	MAE	RMSE
None	Unfitted, primary	0.247	0.279
Endpoint	One constant	0.157	0.210
Molecule	One constant	0.158	0.211
Study	One constant	0.182	0.232
Endpoint	Separate step constants	0.106	0.166
Molecule	Separate step constants	0.106	0.166
Study	Separate step constants	0.124	0.177

N-ethylphenothiazine, N-methylphthalimide and 1,4-benzoquinone cancel the computed ferrocene term algebraically. Each diagnostic excludes every endpoint of its anchor molecule. The resulting error includes the organic anchor's own model and measurement errors. Agreement after reference cancellation therefore does not uniquely identify a ferrocene error.

The spin-only partition includes multiplicity but omits an additional ferrocenium orbital contribution. Matrix spectroscopy and multireference calculations support a degenerate electronic ground state.[1] An idealized additional twofold factor at 298.15 K would lower ferrocenium G by

Table S2: Fc-independent relative-potential diagnostics; all endpoints of each organic anchor are excluded. Errors include the anchor's own model and experimental uncertainty.

Organic anchor	Endpoints	MAE (V)	RMSE (V)
TEMPO	38	0.152	0.217
N-Ethylphenothiazine	38	0.152	0.219
N-Methylphthalimide	38	0.155	0.208
1,4-Benzoquinone	37	0.168	0.250

0.0178 eV, lower the computed Fc absolute potential by 0.0178 V and increase target potentials versus Fc by 0.0178 V. This limiting calculation would worsen the positive first-step bias. It is not an applied correction: solvent, orbital, vibronic and Jahn–Teller splittings are unresolved.

2 Neutral-partner diffuse-basis sensitivity

Five existing benchmark neutral partners were reoptimized with r^{2}SCAN, D4, def2-SVPD and SMD MeCN, with the same active ωB97M-V/def2-TZVPD electronic-energy protocol and a separately checked SVPD Hessian. Each passed stationarity, spin, finite/symmetry, two-column/two-step derivative and minimum checks. A basis-independent certified qRRHO integrator and an independent partition reconstruction were used. No primary state, protocol label or certificate was replaced. Table S3 separates the change in nuclear thermal contribution from the active-energy change due to reoptimization. The existing reduced state and Fc reference are held fixed; only the neutral-partner contribution to the first potential is tested. These are combined reoptimization/Hessian sensitivities; thermal changes are not isolated fixed-geometry basis effects, and conformational relaxation can contribute.

Table S3: SVPD minus primary neutral-partner contributions, in meV; the total also equals the first-potential shift in mV.

Neutral partner	Nuclear	Geometry/SP	Total
1,4-Benzoquinone	-11.29	-0.39	-11.67
1,2-Naphthoquinone	-13.06	-2.23	-15.29
9,10-Phenanthrenequinone	-7.78	-5.66	-13.44
N-Methylphthalimide	-8.63	-5.19	-13.82
N,N'-Diisopropyl-PMDI	-21.27	-13.81	-35.08

3 Exact selected cases: partial assessment with explicit holds

PMDI-ethyl and NDI-ethyl match enumerated exact designs by precursor, product and mapped new bond. All six redox states and the Fc reference were independently reconstructed; their existing grafting evidence hashes are unchanged. The registry now contains exact assessments rather than inherited family passes. Both routes remain held. The exact PMDI mono-N-ethyl/NH precursor is reported in EP0010440B1, Example 19 (printed page 10), with a stated NMR assignment.[2] That access report supports one requirement for PMDI; its selective bead attachment and experimental

redox retention remain unverified. NDI precursor access is still unresolved in the checked collection. The imide attachment precedent establishes only a phthalimide/soluble-polymer analogue. Proposed DBU/MeCN conditions define the computed cycle and do not constitute a demonstrated synthesis protocol. Favorable model thermodynamics therefore do not override the route gate.

Table S4: Joined exact-case evidence. Potentials are V versus computed Fc; grafting free energies are kJ mol^{-1}. Occupancy free energies remain uncomputed for these exact motifs.

Case	E_1	E_2	HCl route	DBU ions	Decision
PMDI-ethyl	-1.076	-1.977	-0.29	-103.07	Hold
NDI-ethyl	-0.764	-1.418	+10.05	-92.73	Hold

These records identify the missing evidence and avoid duplicate full-redox jobs, but do not yet demonstrate a complete enumeration-to-occupancy selection.

3.1 Composition bounds and exact occupancy inputs

For neutral N–H/O–H substitution, the H-capped linear-fragment mass is

$$M = nM_{C_8H_8} + hM_{CHCl} + g(M_{NuH} - M_H - M_{Cl}) + 2M_H + M_{extra}, \quad (1)$$

where n is repeat count, h is the number of original reactive handles and $g \leq h \leq n$ is graft count. The chloromethyl increment replaces a ring hydrogen; an ungrafted handle retains chlorine. Additional inactive mass M_{extra} is explicit and nonnegative. The capacity ceiling for an assumed z electrons per graft is $Q_{upper} = gzF/(3.6M)$. Actual utilization can be lower, including zero. This convention uses neutral dry-fragment mass and does not implicitly include crosslinker, absorbed electrolyte or charging counterions. Charged precursor/attachment classes require a separately declared counterion mass model.

All eight occupancy graphs at $n = 5, h = 3$ reproduce the analytic masses. Positional isomers with the same graft count have the same composition ceiling; their accessibility and interaction energies need not agree. The broader composition grid contains 21 (h, g) cases with one- and two-electron assumptions. No application target is fitted to these values, and the capacity dimension remains unresolved in the decision ledger.

Three exact PMDI inputs, occupied at sites 2, 3 or both 2 and 3, contain 115, 115 and 139 atoms, respectively. They preserve the open ten-carbon backbone, two plain-styrene padding units and any residual benzyl chlorides (Figure S1). Initial structures use seed 20261023, 32 distance-geometry embeddings and up to 2,000 MMFF steps; the best converged initial structure seeds each subsequent search. Connectivity, declared stereochemistry and geometry sanity are checked before calculation. These prepared inputs and running searches are not accepted minima or occupancy free energies. The common empty reference has the same exact graph and configuration as the phthalimide empty control; no occupied phthalimide energy is substituted for a PMDI energy.

4 Relative-basicity controls

The balanced reactions $B + DBUH^+ \rightarrow BH^+ + DBU$ avoid HCl, chloride and a free-proton reference. The exact pyridine, triethylamine and DBU identities match the revised author-maintained MeCN scale, with conjugate-acid pKa values 12.53, 18.83 and 24.31.[3, 4] Four new components and the two reused DBU components pass the same thermodynamic checks and an independent

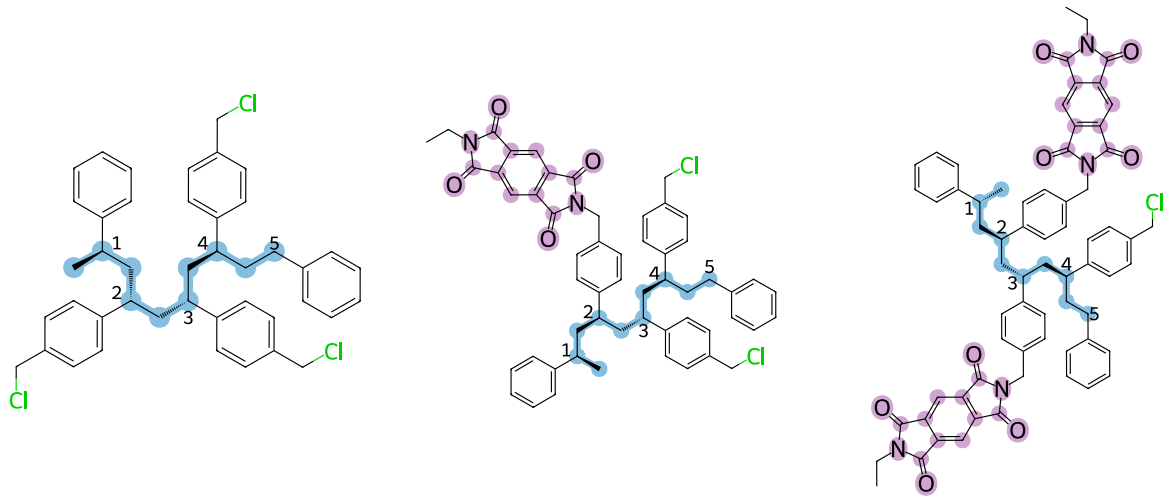


Figure S1: Exact PMDI-ethyl occupancy inputs. Blue highlights the open backbone and purple the diimide cores. Panels show the empty chain, site 2 and sites 2/3; the separate site-3 reference is also prepared. Numbers label repeats. These two-dimensional drawings preserve connectivity and declared stereochemistry, not optimized spatial contact.

six-partition reconstruction. At 298.15 K, computed values are 71.82 and 32.68 kJ mol^{-1}, compared with pKa-derived values 67.24 and 31.28 kJ mol^{-1}; differences are +4.58 and +1.40 kJ mol^{-1}. These values use the exact SI eV-to-molar-energy factor. Historical source records use 96.485 kJ mol^{-1}/eV and are retained without replacement. Source temperature/activity qualifications remain open. This two-reaction control does not validate DIPEA, contact pairs or the complete grafting reaction and cannot assign the DBU/HCl discrepancy to a unique species. The separated model uses bare chloride in a continuum. Recent MeCN ion-solvation comparisons identify both species-dependent errors and sensitivity to absolute single-ion conventions, so the capture discrepancy cannot be assigned to chloride alone.[5] Acid–chloride homoconjugation, such as $HCl + Cl^- \rightleftharpoons HCl_2^-$, would introduce an additional species and acid-dependent speciation. It is not included in either ionic representation; no population or homoconjugation constant is inferred from the free-ion pKa comparison. The source-specific DIPEA note identifies 18.6 as a PCM estimate, although Table 2 of the same author-posted version uses a generic experimental label.[6] We retain that origin conflict and exclude the value from independent measured validation. Holding the computed HCl-reference reactions fixed and using 18.6 with pKa(HCl)=10.30 at the stated comparison temperature gives a capture term of −47.38 kJ mol^{-1}. This changes shared-error cancellation; it is a computational sensitivity, not a calibrated reaction prediction or an accepted uncertainty interval. For methoxy and trimethyl BQ, this gives +10.26/ + 8.66 kJ mol^{-1} compared with the all-DFT DIPEA values −13.34/ − 14.94 kJ mol^{-1}. Neither sign comparison supplies experimental validation.

5 Finite-polymer method controls

Twenty-six of the original twenty-seven electronic oligomer cases currently pass geometry and active-energy checks. The open PMDI trimer has recovered and passed independent force checks. The stacked-start PMDI trimer returned optimizer convergence and saved active energies, but a later GPU-library call terminated abnormally. Separate final-force checks disagree with finite energy differences and remain unresolved, including a subsequent finer-step diagnostic. This case stays outside completed comparisons; no minimum or free energy is inferred from optimizer convergence alone. Construction and XYZ audits verify the intended open-chain topology. A trimer contains six backbone carbons, five backbone C–C bonds and two chain ends, rather than a triangular or cyclic backbone (Figure S2). Proximity of pendant rings does not change that covalent connectivity.

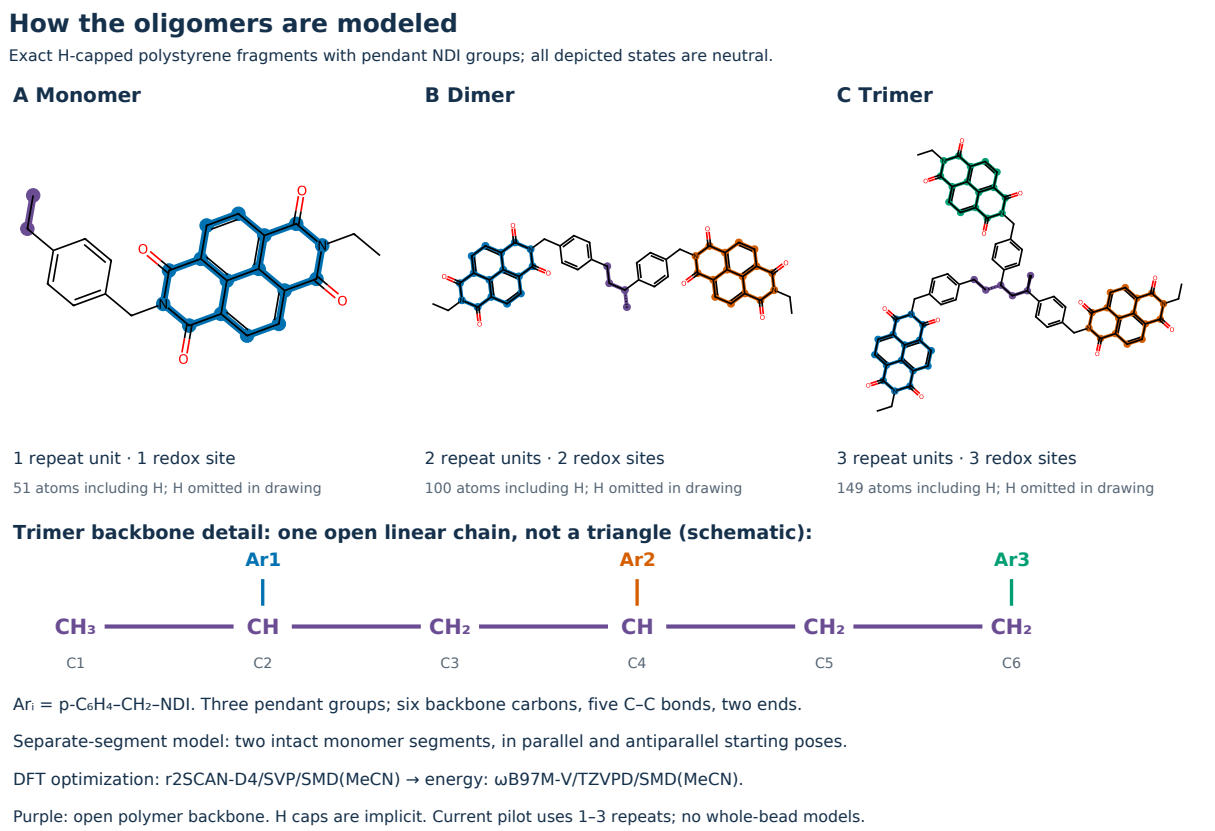


Figure S2: Exact finite-polymer construction used in the interaction controls. The highlighted open backbone distinguishes covalent connectivity from spatial contacts among pendant groups. Monomer, dimer and trimer models are dense local scenarios and do not represent an experimentally measured sequence or graft-loading distribution.

For the six-graph spacing control, all twelve electronic conformers qualify. The selected electronic nonadditivities are -10.98 and -14.50 kJ mol^{-1} for adjacent and one-spacer grafts, respectively. These refer to a matched empty-backbone minimum within the declared, limited sampling set. It demonstrates why a matched empty-backbone reference is necessary; it does not establish a universal distance trend.

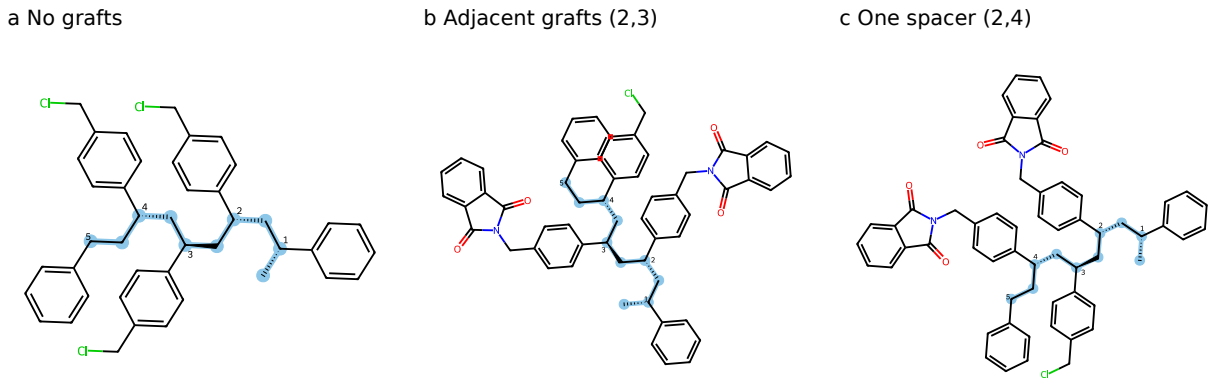


Figure S3: Exact five-repeat phthalimide graphs used in the occupancy cycles. Blue highlights the open ten-carbon backbone. Sites 1 and 5 are plain styrene padding; unoccupied reactive sites retain benzyl chloride. Numbers label repeat positions, and stereobonds retain the declared configuration. These are connectivity drawings, not sampled spatial conformations. Single-occupancy reference graphs are included in the evidence.

All twelve pose-specific native free energies now qualify and have been independently reconstructed from their electronic, Hessian and projected partition records. For a finite set of distinct conformers, we compute

$$G_{\text{finite}} = -RT \ln \sum_{c \in \mathcal{C}} \exp[-G_c/(RT)]. \tag{2}$$

Each distinct optimized minimum has unit occurrence weight; conformer-search hit counts are not degeneracies. Complete-link grouping checks mapped heavy-atom RMSD and electronic-energy agreement. RMSD cutoffs of 0.05, 0.10 and 0.20 Å leave these six pairs distinct, with pairwise RMSDs of 3.50–6.53 Å. The finite partition uses the same temperature, solvent and free-energy protocol for every graph. A partition over a sampled subset does not establish the complete equilibrium ensemble.

The primary adjacent and one-spacer finite-set $\Delta\Delta G$ values are -5.17 and -11.39 kJ mol^{-1}, respectively (Figure S4). Values at rotor cutoffs of 25/50/100 cm^{-1} are $-5.13/-5.17/-4.64$ and $-12.34/-11.39/-11.69$ kJ mol^{-1}. These are partition-model sensitivities, not confidence intervals or bounds on undiscovered-conformer error. Relative to the lowest-G member of each two-pose set, the largest configurational lowering is 0.140 kJ mol^{-1}. The previously reported values, $-5.15/-11.23$ kJ mol^{-1}, used only the minimum-electronic-energy geometry from each graph. Both sets remain tied to the same limited baseline sampling and single backbone configuration.

For DBU and separated ions, the finite-partition free energy of grafting onto an empty backbone is -110.86 kJ mol^{-1} at site 3 and -103.37 kJ mol^{-1} at site 4. With site 2 already grafted, the values become -116.02 and -114.76 kJ mol^{-1}. The contact-pair alternatives are -115.65 to -120.82 kJ mol^{-1} for site 3 and -108.17 to -119.56 kJ mol^{-1} for site 4. Both attachment orders close the same nonadditivity cycle. The base/byproduct contribution cancels from this conditional difference, as does the common empty-chain term when comparing the two spacing cycles. Conformer response in the occupied graphs remains relevant. More favorable subsequent attachment is not automatically desirable when weak unwanted neighboring-site interaction is the objective.

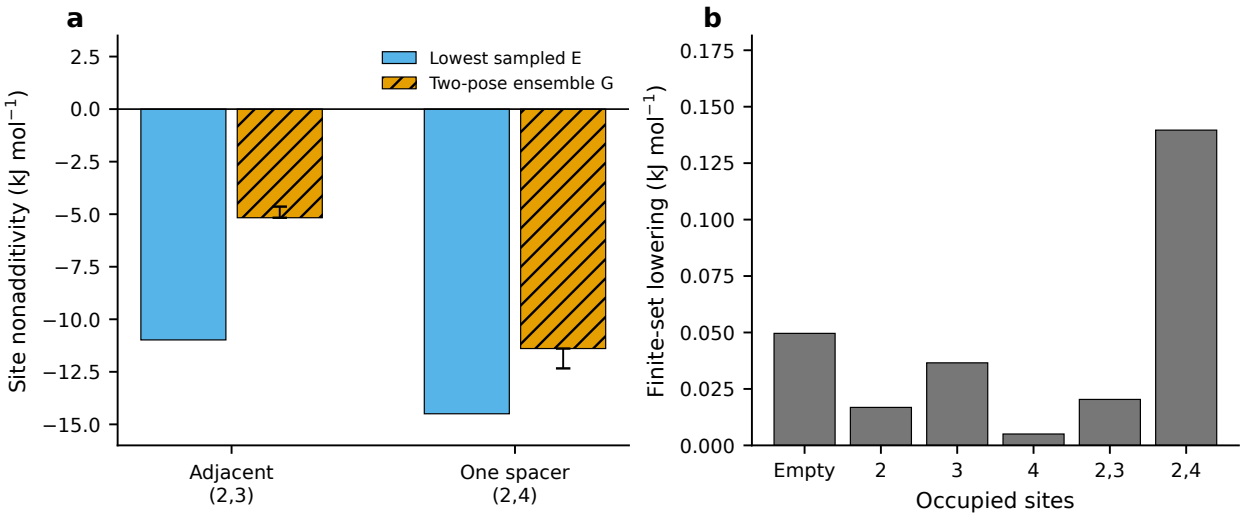


Figure S4: Finite-set phthalimide occupancy control in MeCN at 298.15 K. Panel (a) distinguishes lowest sampled electronic-energy cycles from two-pose free-energy partitions; whiskers vary only the rotor cutoff. Panel (b) shows the configurational lowering within each known two-pose set. Twelve native records pass their checks, but these two-pose sets do not establish converged conformer ensembles or a general spacing trend.

Draft completion item: Complete broader conformer sampling, multiple backbone configurations, DFT re-ranking and exact-motif occupancy assessment before making polymer-design claims. The failed final stacked-trimer force check remains excluded and is not required as a headline result.

The intact-segment decomposition is complete for three sampled pairs using six frozen-fragment single-point calculations (Figure S5). Its partition separates interactions at the complex geometry from deformation relative to relaxed fragments. The interaction energies are -55.25, -44.30 and -50.87 kJ mol^{-1}, with deformation terms $+0.56$, $+0.49$ and $+0.38$ kJ mol^{-1}, respectively. A saved-coordinate audit shows 57–93 optimization steps and heavy-atom input-to-final RMSDs of 0.25–0.70 Å for the complexes, so small deformation sums do not imply negligible structural relaxation. Two totals also contain cancellation: the parallel phthalimide fragments contribute -0.68 and $+1.24$ kJ mol^{-1}, and the PMDI fragments contribute $+1.00$ and -0.62 kJ mol^{-1}. These signed differences refer to one sampled independently optimized monomer at a distinct geometry level; they are not positive-definite strain penalties. Global isolated-monomer minima and Hessian minima are unestablished. The resulting quantities remain electronic, uncorrected for basis-set superposition error and inclusive of backbone and cap contributions. No electron-transfer rate follows from their magnitude.

6 Numerical implementation checks

Diagnostics exposed distinct numerical issues. At a checkpoint of the open PMDI trimer, a legacy solvent cavitation–dispersion–solvent-structure (CDS) force component disagrees with total-energy finite differences by approximately 1.81×10^{-4} hartree bohr^{-1} at the tested coordinate on both integration grids. This discrepancy is not resolved by electronic-grid refinement. Conservative CDS energy derivatives reproduce the tested total-energy differences; their step consistency is checked

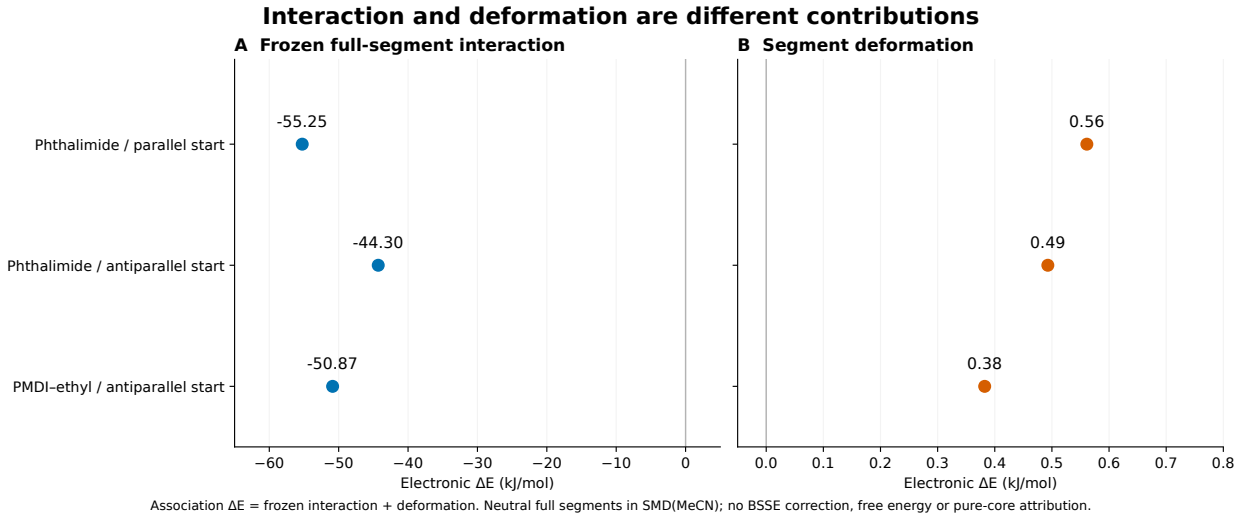


Figure S5: Electronic interaction/deformation partition for three intact segment pairs. Frozen fragments retain the complex geometry without bond cutting or new caps. The partition closes the association-energy cycle but is not a pure pendant-core interaction, a counterpoise-corrected energy or an association free energy.

before use in a continuation. Adaptive refinement of unresolved CDS derivative components recovers the open-trimer optimization. At the returned geometry, independent total-energy derivatives at two step sizes and two tested coordinates agree within 4.2–5.0 $\times 10^{-7}$ hartree bohr^{-1}, below the unchanged 10^{-6} check threshold. This establishes an admitted stationary electronic geometry; no native-Hessian minimum or association free energy is inferred for that original oligomer case.

For the largest spacing Hessian, the original CDS finite-difference step produces a total asymmetry of 2.06×10^{-6}, exceeding the fixed 10^{-6} admission threshold. A separately qualified finer step reduces this to 2.05×10^{-8}. Additive reconstruction replays complete singlet and radical recomputations with maximum matrix differences of 1.33×10^{-11} and 2.83×10^{-9} atomic units, and thermal differences below 1.2×10^{-8} eV. The recovered target then passes the ordinary native gates, with a lowest internal frequency of +7.23 cm^{-1}. These are execution-equivalence results, not a relaxation of physical QC or a high-level validation of grafting energetics.

7 Execution and restart controls

A synthetic control packs one million distinct tasks into one thousand chunks in 75.9 s with a measured preparation peak RSS of 67.2 MiB. Eight workers on two hosts execute the synthetic tasks over a 9.58 s span, excluding packing and startup. Independent arithmetic checks find zero discrepancies. This timing characterizes scheduling and storage, not chemical enumeration, conformer search or DFT throughput.

Separately, all 837 molecular descriptor outputs agree with serial definitions. In the shared-filesystem recovery experiment, 32 real design recipes deduplicate to 17 exact molecular tasks. A worker on one host exits abruptly after four checkpointed tasks; owner verification on that host precedes recovery. A second host completes the remaining thirteen tasks without repeating the four checked tasks. All resumed outputs match uninterrupted execution and a separate serial replay.

This experiment tests descriptor-stage recovery. A distinct quantum interruption experiment is described below; neither establishes million-molecule quantum throughput. A seeded randomized concurrency experiment executes the same 837 exact designs fifteen times, with three repeats per condition and eight tasks per chunk. All 12,555 molecular comparisons match the independent serial definition. Median launch-to-observed completion times are 4.436, 3.147, 2.680 and 2.497 s for one, two, four and eight CPU workers on one host; eight workers across two hosts require 2.585 s (Figure S6). These times include node preflight, SSH, scorer initialization, execution and at most 0.1 s of observation polling; preparation is separate. All requested workers finish, but only seven of eight execute chunks in one single-host repeat. The diminishing returns of this small library do not determine an optimal concurrency for larger or more expensive stages.

A separate sparse Git checkout and newly installed Python 3.11 CPU environment reproduce the declared 32-recipe / 17-design example with system and user site packages disabled. The project loads from that checkout and scientific dependencies load from the new environment. Serial descriptor replay, exact-route holds and the regenerated conditional-grafting table all match; rendered RGB pixels of the figure are identical. This qualifies a small CPU reproduction using curated scientific data. It does not recompute the underlying quantum energies or qualify the complete CUDA environment.

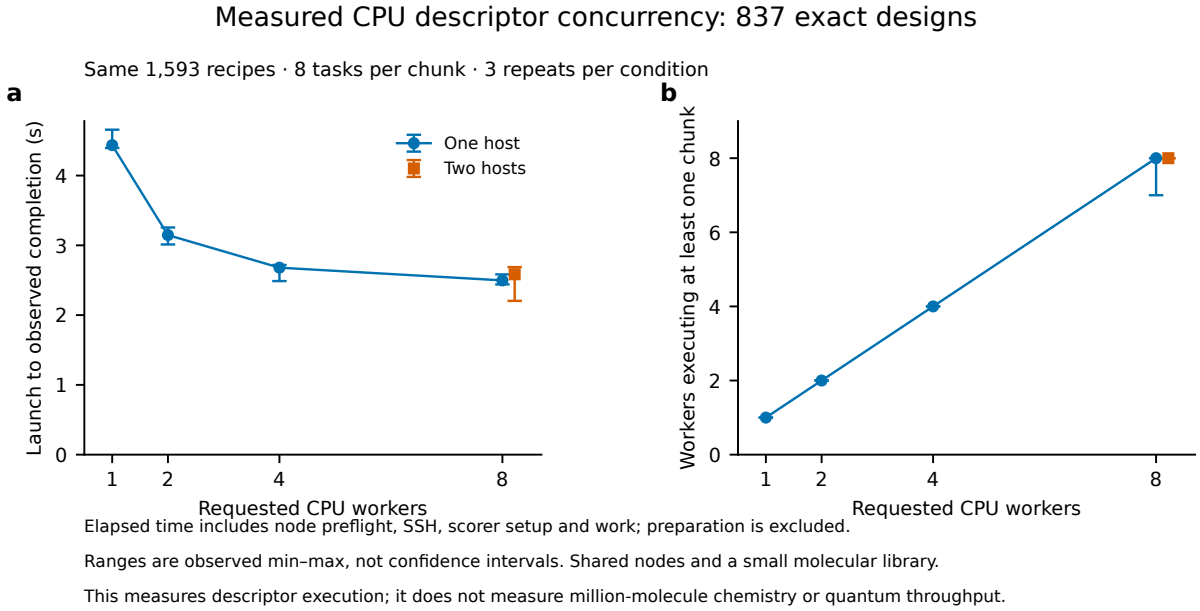


Figure S6: Measured CPU descriptor concurrency for the same 837 exact designs. Three randomized repeats per condition retain every result. Ranges are observed minima and maxima, not confidence intervals. Launch, preflight and scorer overhead are included; preparation is separate. Requested workers and workers executing at least one chunk are distinct. This shared-node, small-library experiment does not measure million-molecule chemistry or quantum throughput.

Draft completion item: Complete conformer-stage adapters, broader quantum recovery controls and accessible archival packaging. Keep stage-specific throughput separate from scientific accuracy.

7.1 Actual quantum interruption and recovery

Two rigid controls, formaldehyde (singlet) and methyl radical (doublet), compare uninterrupted calculations with a fresh process resumed after an intentional exit following three durable gradient evaluations. The exit code is 87; process and start identity confirm termination before resubmission. The saved geometry is unqualified until optimization, active electronic energy, DFT Hessian, projected partition, topology and spin checks pass normally. Acceptance tolerances were 10^{-5} eV for electronic energy, 10^{-4} eV for free energy, 10^{-3} Å aligned RMSD and 0.5 cm^{-1} for internal frequencies. Singlet differences are 4.03×10^{-6} eV, 1.26×10^{-5} eV, 6.87×10^{-6} Å and 0.0685 cm^{-1}; doublet differences are 4.00×10^{-6} eV, 1.11×10^{-6} eV, 3.45×10^{-5} Å and 0.0264 cm^{-1}. Four final partitions were independently reconstructed. This verifies restart equivalence for these two controls, not flexible-polymer sampling or chemical accuracy.

8 Reproducibility package

A locally prepared reviewer package contains 243 thermochemical state records and 32 electronic-only records, together with versioned code, exact conda artifacts, pip distribution versions and an immutable evidence snapshot. Its 2,498 archived files are hash-verified. After relocation, a standard-library interpreter running in isolation reconstructs all 39 benchmark endpoints, 67 HCl-reference grafting values, 20 base-assisted values, two relative-basicity controls and six finite graph partitions. The auxiliary replay covers five alternative-basis controls, two interruption/recovery comparisons and three electronic fragment-decomposition cycles. A separate CPU reconstruction recovers internal modes and nuclear partitions for 242 non-monatomic states from their stored DFT Hessians; it does not load the xTB library or repeat the original physical minimum checks. Seven changed-data, identity, balance, path and unsupported-promotion tests are rejected.

The package represents the preceding numerical evidence epoch, `2026-10-23_all_axes_checked`; the subsequent DIPEA source-origin review and TOC artwork are frozen separately in the present manuscript snapshot. Package replay establishes numerical traceability within these scopes, rather than new electronic-structure calculations, experimental accuracy or broader conformer convergence. No public DOI or deposition is claimed. Full GPU installation/restoration, remaining execution-diagnostic records and final deposition metadata remain open.

9 Evidence terminology

Table S5: Terms used in evidence files and their interpretation.

Term	Meaning
Exact design	A specified precursor/product graph and mapped attachment bond; generating recipes can deduplicate to the same design.
Checked or qualified	The checks required for the named field passed; this is not experimental validation.
Native free energy	Active electronic energy plus the projected geometry-level DFT nuclear partition and declared spin partition.
Frozen snapshot	An immutable, hash-verified copy of specified evidence; newer live results require a new snapshot.
Proceed / hold / exclude	Route decisions within the stated evidence scope; hold means required exact-route evidence is incomplete.
Endpoint	One sourced redox-potential comparison, with exact molecular identity, state pair and reference convention.
Pose	One conformational starting geometry; the final optimized contact class is checked separately.

References

[1] Jan Fulara, Karol Filipkowski, and John P. Maier. Electronic Transition of Ferrocenium: Neon Matrix and CASPT2 Studies. *The Journal of Physical Chemistry C*, 121:10694–10697, 2017. doi: 10.1021/acs.jpcc.6b10391.

[2] Richard J. Bochis, Michael H. Fisher, and Bruce O. Linn. Pyromellitic diimides, processes for their preparation, these compounds in orally administrable form for use in increasing the feed efficiency of a ruminant animal and compositions for this purpose containing them. European patent EP0010440B1, published January 26, 1983, 1983. URL https://patents.google.com/patent/EP0010440B1/en. Example 19, printed p. 10; precursor identity only.

[3] Sofja Tshepelevitsh, Agnes Kütt, Märt Lõkov, Ivari Kaljurand, Jaan Saame, Agnes Heering, Paul G. Plieger, Robert Vianello, and Ivo Leito. On the basicity of organic bases in different media. *European Journal of Organic Chemistry*, 2019:6735–6748, 2019. doi: 10.1002/ejoc.201900956.

[4] University of Tartu, Chair of Analytical Chemistry. Author-maintained basicity collection and recommended acid table. https://analytical.chem.ut.ee/databases/pka-values/. Accessed 22 October 2026. Acid table: https://analytical.chem.ut.ee/HA_UT/pKa_Table_Water_Acetonitrile_DCE_Gas_Phase.pdf.

[5] Josefredo R. Pliego, Jr. Hybrid cluster-continuum method for single-ion solvation free energy in acetonitrile solvent. *The Journal of Physical Chemistry A*, 128:6440–6449, 2024. doi: 10.1021/acs.jpca.4c03593.

[6] David J. Heldebrant, Phillip K. Koech, James E. Rainbolt, Feng Zheng, Tricia Smurthwaite, Charles J. Freeman, Merit Oss, and Ivo Leito. Performance of single-component CO2-binding organic liquids (CO2BOLs) for post combustion CO2 capture. *Chemical Engineering Journal*, 171(3):794–800, 2011. doi: 10.1016/j.cej.2011.02.012.